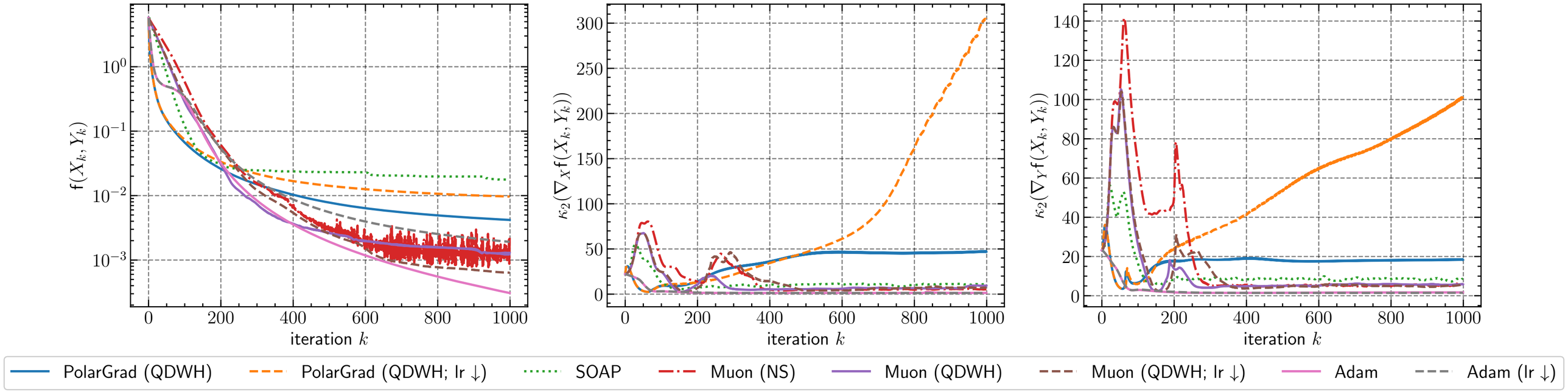
$f(X_k, Y_k)$
10^0
10^{-1}
10^{-2}
10^{-3}
0
200
400
600
800
1000
iteration k
$\kappa_2(\nabla_X f(X_k, Y_k))$
300
250
200
150
100
50
0
iteration k
$\kappa_2(\nabla_Y f(X_k, Y_k))$
140
120
100
80
60
40
20
0
iteration k
PolarGrad (QDWH)
PolarGrad (QDWH; lr ↓)
SOAP
Muon (NS)
Muon (QDWH)
Muon (QDWH; lr ↓)
Adam
Adam (lr ↓)